
Advanced Policy Gradient Reinforcement Learning

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AI Masters

March 30, 2026

Bias-Variance trade-off

Reminder: Policy Gradient

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Critic: an approximation $V_\phi(s) \approx V^\pi(s)$.

Important: it is a **biased** estimation.

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$Q^{\pi_\theta}(s_t, a_t) \approx y_Q = r + \gamma V_\phi(s_{t+1})$ is also biased.

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For Actor:

$$\nabla := \nabla_{\theta} \log \pi_{\theta}(a \mid s) \underbrace{\Psi(s, a)}_{\substack{\text{advantage} \\ \text{estimator}}}$$

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N -step	$\Psi_{(N)}(s, a) := r + \gamma r' + \dots + \gamma^N V(s^{(N)}) - V(s)$	intermediate	intermediate
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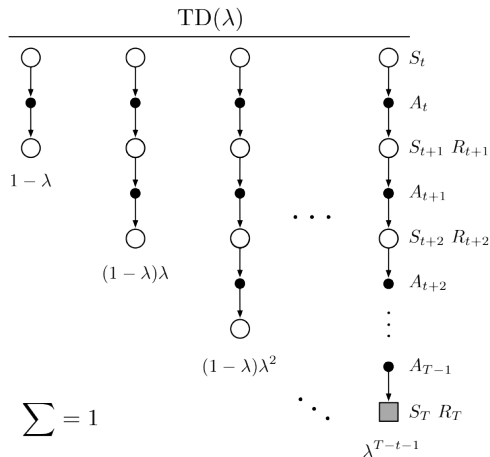
Problem: hard to choose N .

Reminder: TD(λ)



Can't choose? Use them all.

Estimation	Weight
$\Psi_{(1)}(s, a)$	$1 - \lambda$
$\Psi_{(2)}(s, a)$	$(1 - \lambda)\lambda$
$\Psi_{(3)}(s, a)$	$(1 - \lambda)\lambda^2$
$\vdots$	$\vdots$



Reminder: TD(λ)

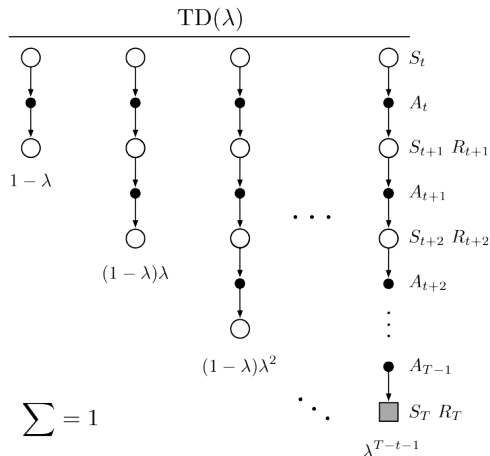


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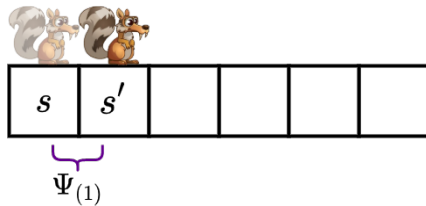
Generalized Advantage Estimation:

$$\Psi_{\text{GAE}}(s, a) = (1 - \lambda) \sum_{N=1}^{\infty} \lambda^{N-1} \Psi_{(N)}(s, a)$$



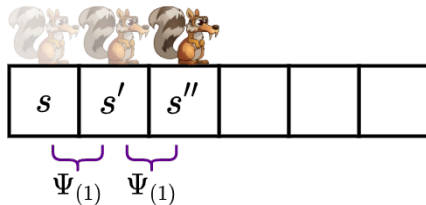
Discounted sum of TD-errors

$$\overbrace{r + \gamma V(s') - V(s)}^{\Psi_{(1)}(s,a)} +$$



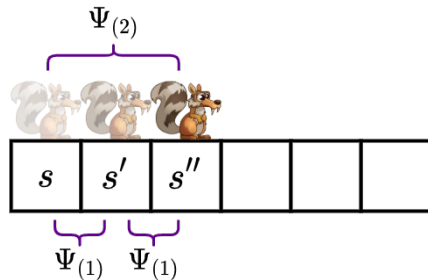
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$$\overbrace{r + \gamma V(s') - V(s)}^{\Psi_{(1)}(s,a)} + \overbrace{\gamma r' + \gamma^2 V(s'') - \gamma V(s')}^{\gamma \Psi_{(1)}(s',a')}$$



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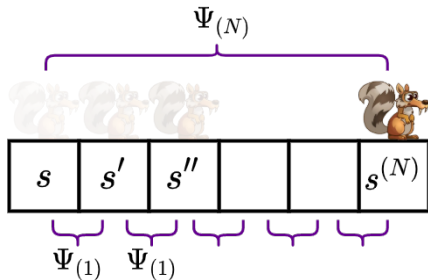


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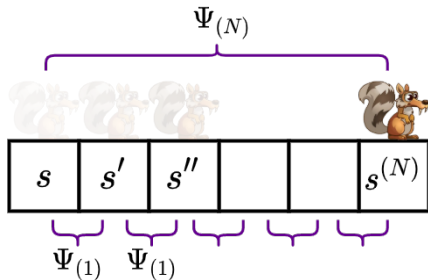


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Rule of thumb: discounted sum of discounted sums is discounted sum =D

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Equivalent form of GAE

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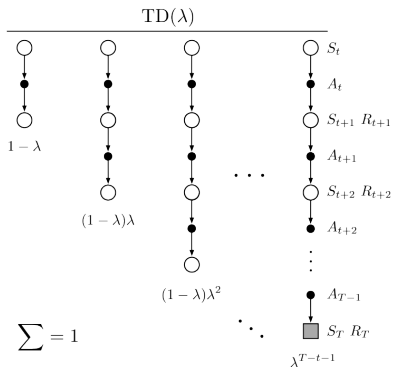
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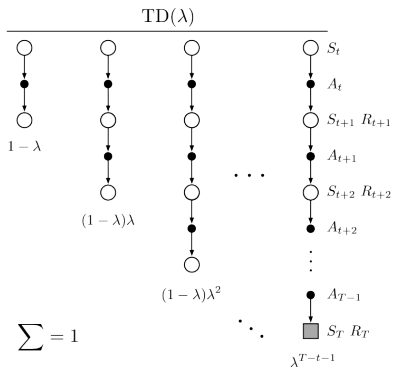


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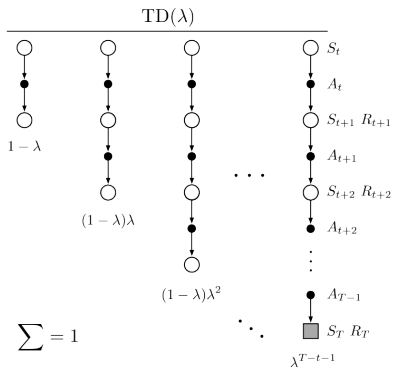
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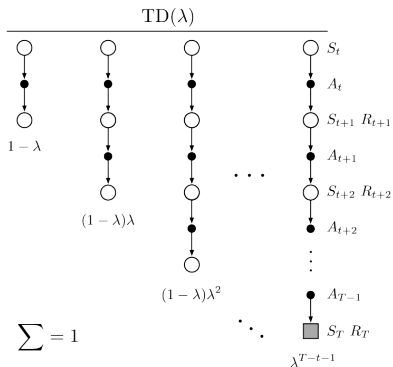
Equation used in practice:

$$\begin{aligned} \psi^{\text{GAE}}(s_t, a_t) = & \psi_{(1)}(s_t, a_t) + \\ & + \lambda \gamma (1 - \text{done}_{t+1}) \psi^{\text{GAE}}(s_{t+1}, a_{t+1}) \end{aligned}$$

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Question: Can we have Monte Carlo estimation of $\mathbb{E}_{\mathcal{T} \sim \pi_\theta} \sum_{t \geq 0} (\cdot)$ without playing full episodes?

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where $d_\pi(s)$ is **state visitation frequency** of policy π .

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But **sequential states from one rollout are correlated**.

$$\nabla_\theta J(\pi_\theta) = \frac{1}{1 - \gamma} \underbrace{\mathbb{E}_{s \sim d_{\pi_\theta}(s)} \mathbb{E}_{a \sim \pi_\theta(a|s)}}_{\text{data generated by } \pi_\theta} \nabla_\theta \log \pi_\theta(a | s) A^{\pi_\theta}(s, a)$$

(sample pairs s, a from trajectories $\mathcal{T} \sim \pi_\theta$)

GAE in Advantage Actor-Critic



Longer rollouts produce richer GAE ensemble.

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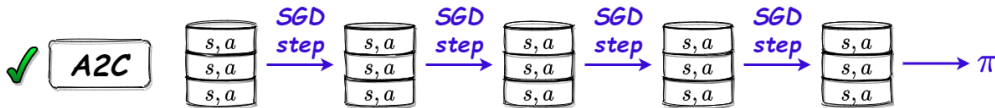
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In A2C rollouts are usually short, so $\lambda = 1$ is common choice.
(sometimes called **max-trace** estimation)

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Value-based (DQN+)

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 trains $Q^*(s, a)$;
(complicated intermediate stage)

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 trains policy directly;
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 off-policy;
(can use experience replay)

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 on-policy;
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Proximal Policy Optimization (PPO)

Off-policy Policy Gradient?

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- want to optimize π_{θ} (compute gradient for current θ);

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May be we can somehow express $J(\pi_{\theta})$ using $A^{\pi^{\text{old}}}(s, a)$?

Connecting two policies



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We will utilize the following trick (**telescoping sums**): for any sequence a_t , s.t. $\lim_{t \rightarrow \infty} a_t = 0$:

$$\sum_{t \geq 0}^{\infty} (a_{t+1} - a_t) = -a_0$$

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Let's perform **reward shaping** using another policy value function!

We will utilize the following trick (**telescoping sums**): for any sequence a_t , s.t. $\lim_{t \rightarrow \infty} a_t = 0$:

$$\sum_{t \geq 0}^{\infty} (a_{t+1} - a_t) = -a_0$$

So, for arbitrary trajectory $\mathcal{T} := s_0, a_0, s_1, a_1, \dots$ and arbitrary policy π :

$$-V^{\pi}(s_0) = \sum_{t \geq 0} [\gamma^{t+1} V^{\pi}(s_{t+1}) - \gamma^t V^{\pi}(s_t)] \quad (1)$$

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We can switch our optimization function to:

$$J(\pi_\theta) - J(\pi^{\text{old}}) = \frac{1}{1-\gamma} \underbrace{\mathbb{E}_{s \sim d_{\pi_\theta}(s)} \mathbb{E}_{a \sim \pi_\theta(a|s)}}_{\substack{\text{collect data with } \pi_\theta \\ \text{(wrong! we don't have it!)}}} \overbrace{A^{\pi^{\text{old}}}(s, a)}^{\substack{\text{old critic!} \\ \text{(good: can train it!)}}}$$

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 - ▶ optimizing θ with fixed *data* (fixed $\Psi(s, a) \approx A^{\pi^{\text{old}}}(s, a)$) will learn $\pi_\theta(a|s) = 1$ if $\Psi(s, a) > 0$, $\pi_\theta(a|s) = 0$ otherwise.

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Minorization-maximization algorithm



We discovered a **variational lower bound** for our objective:

$$J(\pi_\theta) - J(\pi^{\text{old}}) \geq L_{\pi^{\text{old}}}(\theta) - C \text{KL}^{\text{max}}(\pi^{\text{old}} \parallel \pi_\theta)$$

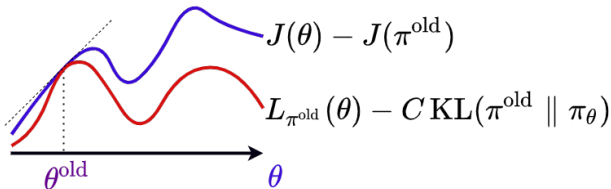
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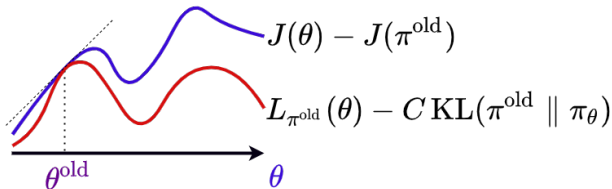
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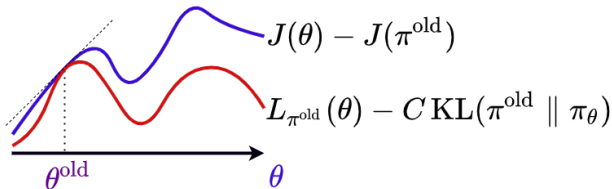
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✓ guarantees monotonic improvement!



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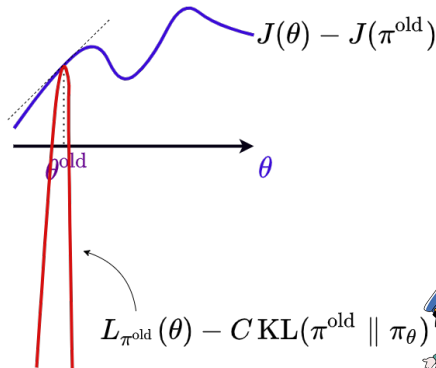


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X impractical: real C is too huge!



Trust-Region optimization



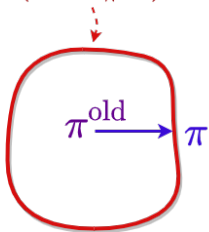
Optimize $L_{\pi^{\text{old}}}(\theta)$, but guarantee that π_θ does not change too much from π^{old} !

Trust-Region optimization



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$$\text{KL}(\pi^{\text{old}} \parallel \pi) \leq \delta$$



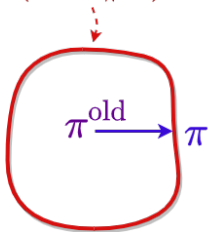
$$\begin{cases} L_{\pi^{\text{old}}}(\theta) \rightarrow \max_{\theta} \\ \mathbb{E}_{s \sim d_{\pi^{\text{old}}}(s)} \text{KL}(\pi^{\text{old}}(\cdot | s) \parallel \pi_{\theta}(\cdot | s)) \leq \delta \end{cases}$$

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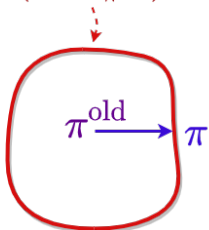
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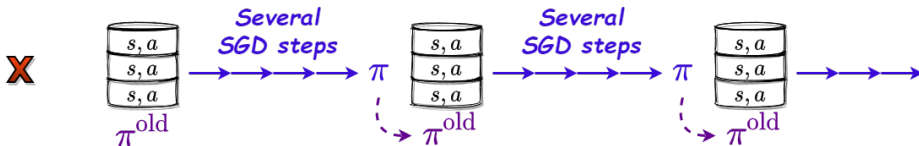
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Algorithm **TRPO** attempts to solve this conditioned task approximately. *Complicated implementation.*

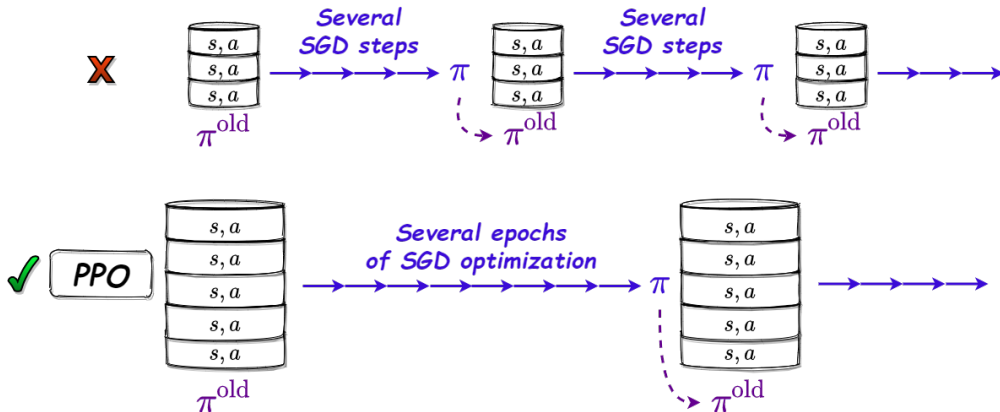
Proximal Policy Optimization (PPO): Pipeline

$$\mathbb{E}_{s \sim d_{\pi^{\text{old}}}(s)} \mathbb{E}_{a \sim \pi^{\text{old}}(a|s)} \frac{\pi_{\theta}(a|s)}{\pi^{\text{old}}(a|s)} A^{\pi^{\text{old}}}(s, a) - C \text{KL}(\pi^{\text{old}} \parallel \pi_{\theta}) \rightarrow \max_{\theta}$$



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Clipping Objective

$$L_{\pi^{\text{old}}}(\theta) - C \text{KL}(\pi^{\text{old}} \parallel \pi_{\theta}) \rightarrow \max_{\theta}$$

Default surrogate function:

$$\rho(\theta) := \frac{\pi_{\theta}(a \mid s)}{\pi^{\text{old}}(a \mid s)}$$

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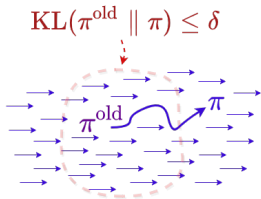
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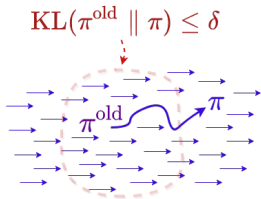
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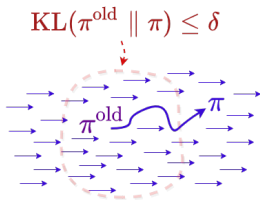
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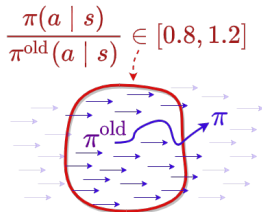
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Recalling lower bound intuition

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Advantage Sign	Direction	Bad ratio case	Gradient
$A^{\pi^{\text{old}}}(s, a) \geq 0$			

Recalling lower bound intuition

$$\mathbb{E}_{s \sim d_{\pi^{\text{old}}}(s)} \mathbb{E}_{a \sim \pi^{\text{old}}(a|s)} \underbrace{\min(\rho(\theta) A^{\pi^{\text{old}}}(s, a), \quad \overbrace{\rho^{\text{clip}}(\theta) A^{\pi^{\text{old}}}(s, a)}^{\text{term with clipped importance sampling weight}})}_{\text{original term}} \underbrace{) - C \text{KL}(\pi^{\text{old}} \parallel \pi_{\theta})}_{\text{«regularization»}} \rightarrow \max_{\theta}$$

Advantage Sign	Direction	Bad ratio case	Gradient
$A^{\pi^{\text{old}}}(s, a) \geq 0$	$\pi_{\theta}(a s) \uparrow$		

Recalling lower bound intuition

$$\mathbb{E}_{s \sim d_{\pi^{\text{old}}}(s)} \mathbb{E}_{a \sim \pi^{\text{old}}(a|s)} \underbrace{\min(\rho(\theta) A^{\pi^{\text{old}}}(s, a), \quad \overbrace{\rho^{\text{clip}}(\theta) A^{\pi^{\text{old}}}(s, a)}^{\text{term with clipped importance sampling weight}})}_{\text{original term}} - \underbrace{C \text{KL}(\pi^{\text{old}} \parallel \pi_{\theta})}_{\text{«regularization»}} \rightarrow \max_{\theta}$$

Advantage Sign	Direction	Bad ratio case	Gradient
$A^{\pi^{\text{old}}}(s, a) \geq 0$	$\pi_{\theta}(a s) \uparrow$	$\rho(\theta) > 1.2$	

Recalling lower bound intuition

$$\mathbb{E}_{s \sim d_{\pi^{\text{old}}}(s)} \mathbb{E}_{a \sim \pi^{\text{old}}(a|s)} \min(\overbrace{\rho(\theta) A^{\pi^{\text{old}}}(s, a)}^{\text{original term}}, \overbrace{\rho^{\text{clip}}(\theta) A^{\pi^{\text{old}}}(s, a)}^{\text{term with clipped importance sampling weight}}) - \overbrace{C \text{KL}(\pi^{\text{old}} \parallel \pi_{\theta})}^{\text{«regularization»}} \rightarrow \max_{\theta}$$

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Advantage Sign	Direction	Bad ratio case	Gradient
$A^{\pi^{\text{old}}}(s, a) \geq 0$	$\pi_{\theta}(a s) \uparrow$	$\rho(\theta) > 1.2$ $\rho(\theta) < 0.8$	0

Recalling lower bound intuition

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Recalling lower bound intuition

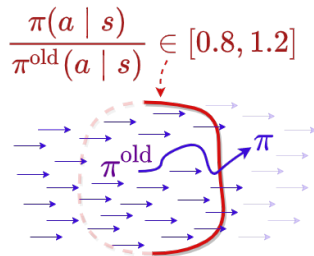
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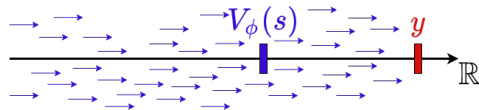
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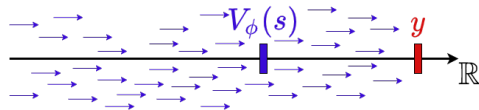
Clipped Critic Loss

$$\text{Loss}(\phi) := (y - V(\phi))^2 =$$



Clipped Critic Loss

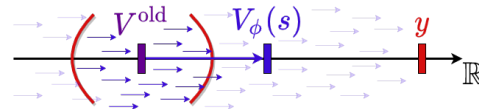
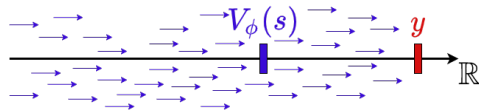
$$\begin{aligned}\text{Loss}(\phi) &:= (y - V(\phi))^2 = \\ &= (y - V^{\text{old}} + V^{\text{old}} - V(\phi))^2\end{aligned}$$



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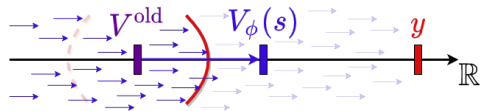
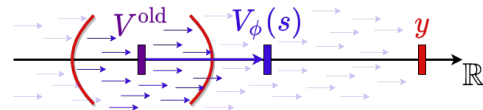
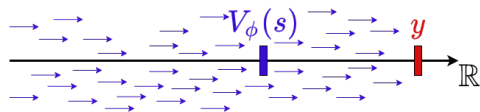


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$$\max(\text{Loss}(\phi), \text{Loss}^{\text{clip}}(\phi))$$



Proximal Policy Optimization: implementation matters

Key elements:

- ✓ Clipped policy loss
- ✓ Clipped critic loss
- ✓ GAE

Pipeline details:

- ! Advantage normalization in mini-batches
- No KL regularization
- Entropy loss

¹divided by running std of collected cumulative rewards

²can be critical in continuous control

Other hacks:

- ! Reward normalization¹ and clipping
- Observations normalization and clipping²
- Orthogonal initialization of layers
- ϵ (clipping parameter) annealing

Standard tricks:

- Adam, learning rate annealing
- Tanh activation functions
- ! Gradient clipping

Full Pipeline: pt.I

Proximal Policy Optimization (PPO)

Initialize $\pi(a \mid s, \theta), V_{\phi}(s);$

Full Pipeline: pt.I

Proximal Policy Optimization (PPO)

Initialize $\pi(a \mid s, \theta), V_\phi(s);$

for $k = 0, 1, 2 \dots$

- collect several rollouts $s_0, a_0, r_0, s_1, \text{done}_1, a_1 \dots s_N, \text{done}_N$ using $\pi(a \mid s, \theta);$
store probabilities of selected actions as $\pi^{\text{old}}(a_t \mid s_t) := \pi(a_t \mid s_t, \theta)$
store critic output as $V^{\text{old}}(s_t) := V_\phi(s_t)$

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- for t from $N - 2$ to 0 :
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- compute critic targets: $y(s_t) := \Psi^{\text{GAE}}(s_t, a_t) + V_\phi(s_t)$
- construct dataset of $(s_t, a_t, \Psi^{\text{GAE}}(s_t, a_t), y(s_t), \pi^{\text{old}}(a_t \mid s_t), V^{\text{old}}(s_t))$

Full Pipeline: pt.II

Proximal Policy Optimization (PPO) -- cont.

- go through dataset n_epochs times, sampling mini-batches of size B ; for each mini-batch:

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$$\rho(s, a, \theta) := \frac{\pi(a \mid s, \theta)}{\pi^{\text{old}}(a \mid s)}, \quad \rho^{\text{clip}}(s, a, \theta) = \text{clip}(\rho(s, a, \theta), 1 - \epsilon, 1 + \epsilon)$$

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- ▶ update actor:

$$L_1(s, a, \theta) := \rho(s, a, \theta) \Psi^{GAE}(s, a), \quad L_2(s, a, \theta) := \rho^{clip}(s, a, \theta) \Psi^{GAE}(s, a)$$

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} \frac{1}{B} \sum_{s, a} \min(L_1(s, a, \theta), L_2(s, a, \theta))$$

Full Pipeline: pt.II

Proximal Policy Optimization (PPO) -- cont.

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$$\theta \leftarrow \theta + \alpha \nabla_{\theta} \frac{1}{B} \sum_{s, a} \min(L_1(s, a, \theta), L_2(s, a, \theta))$$

- update critic:

$$\text{Loss}_1(s, \phi) := (y(s) - V_{\phi}(s))^2$$

$$\text{Loss}_2(s, \phi) := \left(y(s) - V^{old}(s) - \text{clip}(V_{\phi}(s) - V^{old}(s), \hat{\epsilon}, -\hat{\epsilon}) \right)^2$$

$$\phi \leftarrow \phi - \alpha \nabla_{\phi} \frac{1}{B} \sum_s \max(\text{Loss}_1(s, \phi), \text{Loss}_2(s, \phi))$$

Go-to Model Free Algorithms



	<i>Discrete Action Space</i>	<i>Continuous Action Space</i>
<i>On-policy</i>	PPO	PPO
<i>Off-policy</i>	DQN+	TD3 / SAC <i>(next time)</i>

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Hometask 5:

Continuous control; you can choose PPO, TD3 or SAC.



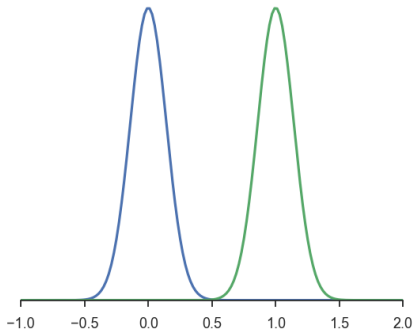
Literature

- High-Dimensional Continuous Control Using Generalized Advantage Estimation;
- Trust Region Policy Optimization;
- Proximal Policy Optimization Algorithms;
- Implementation Matters in Deep Policy Gradients: A Case Study on PPO and TRPO;
- OpenAI Five;

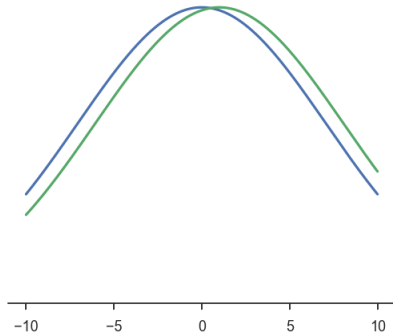
Appendix: Trust-Region Policy Optimization (TRPO)

Parameter space vs distribution space

Two gaussians: $\mathcal{N}(0, 0.2), \mathcal{N}(1, 0.2)$

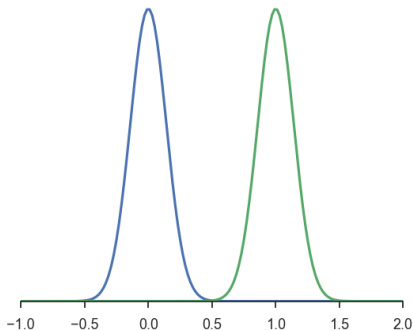


Two gaussians: $\mathcal{N}(0, 10), \mathcal{N}(1, 10)$

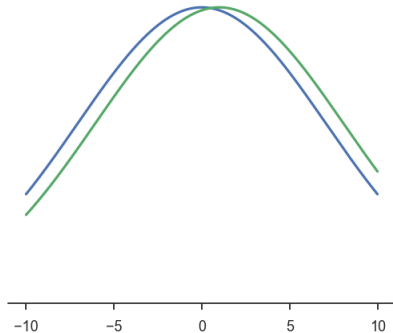


Parameter space vs distribution space

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Two gaussians: $\mathcal{N}(0, 10), \mathcal{N}(1, 10)$



Euclidean distance: $\sqrt{(\mu_1 - \mu_2)^2 + (\sigma_1 - \sigma_2)^2}$

Statistical distance: e.g. $\text{KL}(\mathcal{N}(\mu_1, \sigma_1) \parallel \mathcal{N}(\mu_2, \sigma_2))$

Step size issue: riding from the cliff



$$\theta_{k+1} = \theta_k + \alpha \hat{g}_k$$

If step size too large:

- step too far leads to bad policy;
- we start *collecting data* with bad policy;
- $\implies$ **performance collapse**

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We need optimization method that measures distance in **distribution space**, not in **parameter space**!

A view on optimization methods

Consider the following optimization task:

$$G(\theta) := G(q_\theta(x)) \rightarrow \min_{\theta}$$

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General scheme of trust region optimization methods

- start with arbitrary θ_0 ;
- **for** $k = 0, 1, 2, \dots$:
 - ▶ construct a **model**, some local approximation of $G(\theta)$ near θ_k :

$$m_k(\theta) \approx G(\theta)$$

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- ▶ select some **trust region** where you assume this model is more or less correct:

$$r(\theta, \theta_k) \leq \delta$$

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- ▶ seek θ_{k+1} as solution of

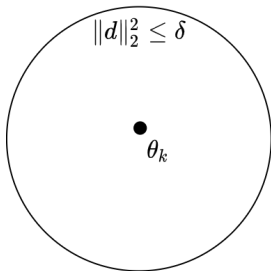
$$\begin{cases} m_k(\theta) \rightarrow \min_{\theta} \\ r(\theta, \theta_k) \leq \delta \end{cases}$$

Natural gradient descent

$$G(\theta) := G(q_\theta(x)) \rightarrow \min_{\theta}$$

Gradient Descent

$$\begin{cases} G(\theta_k + d) \approx G(\theta_k) + \nabla_{\theta} G(\theta_k)^T d \rightarrow \min_d \\ \|d\|_2 \leq \delta \end{cases}$$

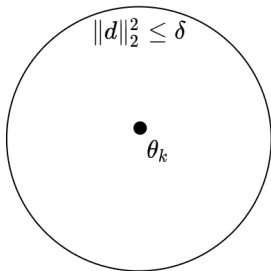


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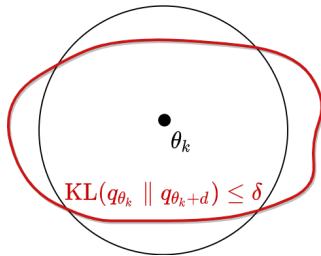
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Natural Gradient Descent

$$\begin{cases} G(\theta_k + d) \approx G(\theta_k) + \nabla_{\theta} G(\theta_k)^T d \rightarrow \min_d \\ \text{KL}(q_{\theta_k}(x) \parallel q_{\theta_k+d}(x)) \leq \delta \end{cases}$$

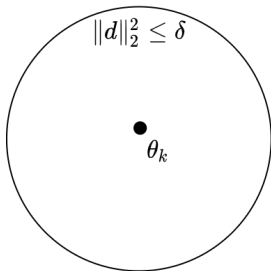


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$$G(\theta) := G(q_\theta(x)) \rightarrow \min_{\theta}$$

Gradient Descent

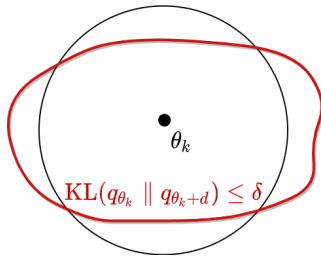
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Solution: $d \propto -\nabla_{\theta} G(\theta_k)$

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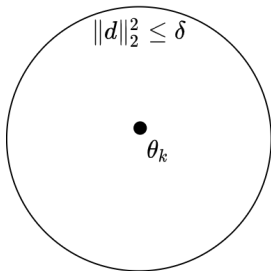


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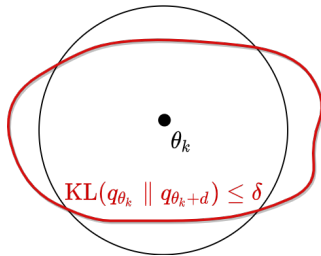
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Solution: ?!?



Use Taylor expansion for
 $\text{KL}(q_{\theta_k}(x) \parallel q_{\theta_k+d}(x)) \leq \delta$

Fisher Matrix



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Let $F(\theta_k)$ be the **Fisher matrix** of $q_\theta(x)$ at point θ_k :

$$F(\theta_k) := -\mathbb{E}_{x \sim q_\theta(x)} \nabla_\theta^2 \log q_\theta(x)$$

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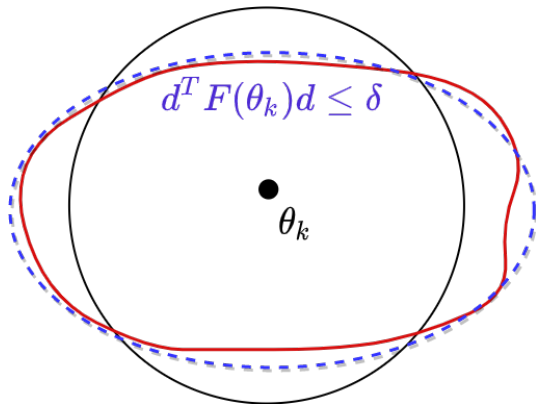
$$F(\theta_k) := -\mathbb{E}_{x \sim q_{\theta_k}(x)} \nabla_{\theta}^2 \log q_{\theta_k}(x)$$

Then Taylor approximation of second order is:

$$\text{KL}(q_{\theta_k}(x) \parallel q_{\theta_k+d}(x)) \approx \frac{1}{2} d^T F(\theta_k) d$$

Direction of natural gradient

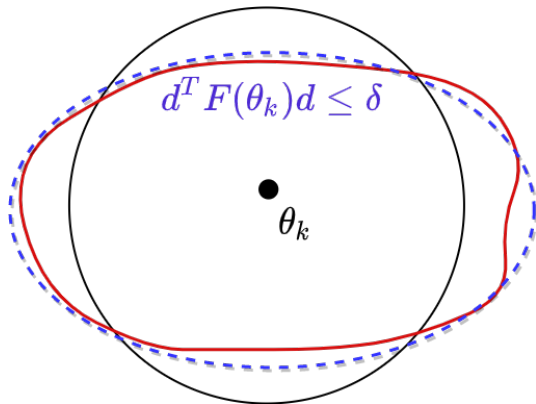
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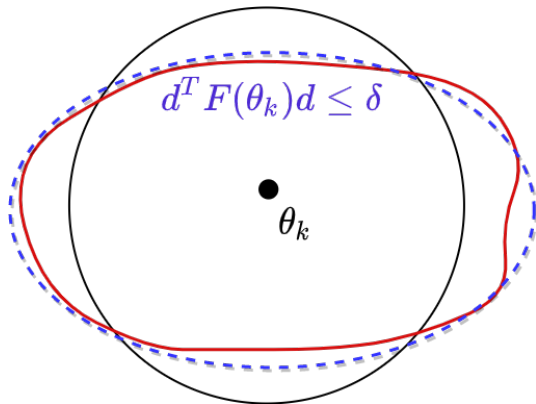
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Natural gradient uses the following update:

$$\theta_{k+1} = \theta_k - \alpha F^{-1}(\theta_k) \nabla_{\theta} G(\theta_k),$$

where α is determined by the choice of δ .



Trust-Region optimization



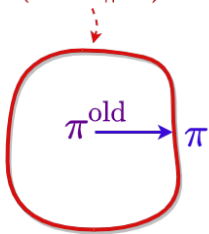
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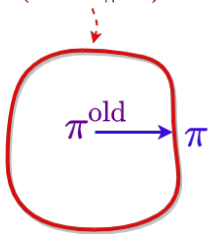
$$\begin{cases} L_{\pi^{\text{old}}}(\theta) \rightarrow \max_{\theta} \\ \mathbb{E}_{s \sim d_{\pi^{\text{old}}}(s)} \text{KL}(\pi^{\text{old}}(\cdot | s) \parallel \pi_{\theta}(\cdot | s)) \leq \delta \end{cases}$$

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- $L_{\pi^{\text{old}}}(\theta)$ is the **model**;
- condition is **trust region** to this model;

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Use Taylor expansion for both model and condition to solve this system!

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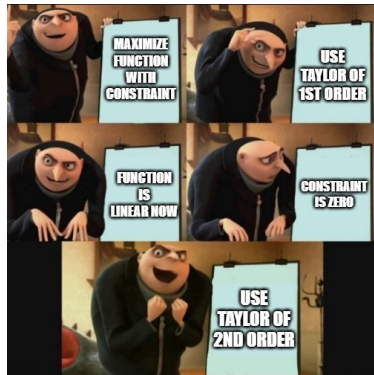
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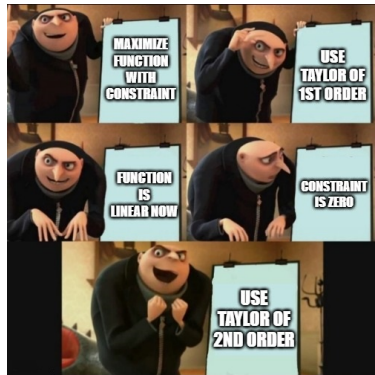


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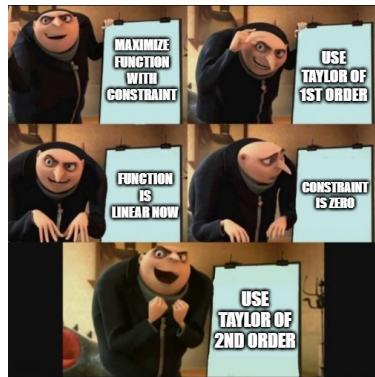
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Natural Gradient Ascent

Solution: $d \propto H^{-1}g$

Reality check: can we compute inverse hessian?

$$\theta_{k+1} = \theta_k + \alpha H^{-1}g$$

Danger !

Natural gradient descent is similar to second-order methods.

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X H is a matrix $h \times h$
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Line Search

How to determine learning rate α_k ?

(it is interconnected with δ in trust region view)

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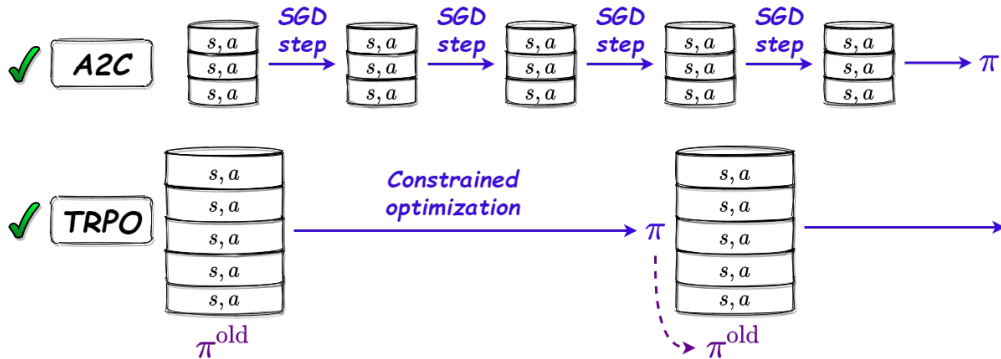
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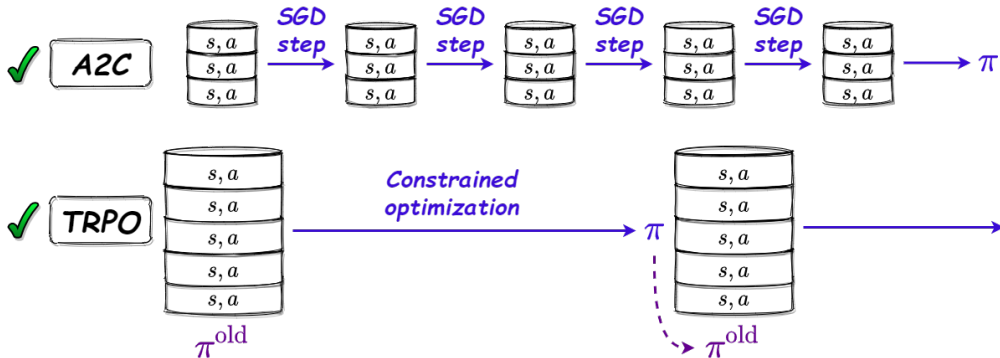
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e. g. decrease α_k in half until these constraints are met.

TRPO: pipeline



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- ✓ better utilization of GAE
- ✓ more sample-efficient than A2C
- ✓ considered quite robust

- ✗ critic has to be trained in separate optimization process
- ✗ computationally expensive
- ✗ complicated :{